

Clinical Validation of Explainable AI in Healthcare Readmission Prediction: A Systematic Gap Analysis

1st Robert Borkar
School of Computing
Dublin City University
Dublin, Ireland
robert.borkar2@mail.dcu.ie

2nd Niket Ahire
School of Computing
Dublin City University
Dublin, Ireland
niketsuresh.ahire2@mail.dcu.ie

3rd Martin Crane
School of Computing
Dublin City University
Dublin, Ireland
martin.crane@dcu.ie

Abstract—Hospital readmissions within 30 days accounted for more than \$26 billion annually in the US[11]. Machine learning models have reached AUROC ≥ 0.75 –0.89 on prediction, but methods for providing explanations such as SHAP and LIME have not been validated in comparison with clinical evidence. Analysis reveals that 74% of XAI healthcare papers output feature rankings but do not provide clinical confirmation that covers their use, which hinders adoption. We identify four gaps: comparative SHAP for architectures, clinical validation frameworks, class imbalance impact on explanation stability, and dimensionality reduction XAI integration.

Index Terms—explainable AI, SHAP, LIME, hospital readmission prediction, clinical validation

I. INTRODUCTION

ML readmission models achieve excellent test characteristics (AUROC 0.75–0.89), but remain black boxes. SHAP and LIME produce feature importance rankings that explain which patient features influence their predictions, but 74% of healthcare XAI research neglects clinical validation [5]–[6]. Practitioners are unable to distinguish signal from statistical noise, limiting clinical adoption.

We analyzed recent XAI applications targeted at readmission prediction and identified four major research gaps: comparative explainability across different architectures, systematic clinical validation standards, the impact of class imbalance on explanation stability, and how to combine dimensionality reduction with explanations.

II. MODEL EVOLUTION AND VALIDATION GAPS

A. Ensemble and Deep Learning Progression

Yu and Xie [1] reached 0.891 recall on 651,816 patient records from Chinese hospitals using their MWBS (Multi-model Weighted Bagging and boosting based on Synthetic minority oversampling) approach. The method combined cost-sensitive boosting with multiple classifiers to handle the 15% readmission imbalance, improving recall by 22.7% over baseline methods. Performance was strong but clinical interpretation was absent—practitioners could not understand which patient characteristics drove individual risk assessments.

Li et al. [2] advanced temporal modeling with bidirectional LSTMs on 21,002 Medicare patients from MIMIC-III. The model processed admission-level sequential EHR data including surgeries, medications, ICD codes, and emergency

department visits, beating the LACE baseline used in clinical practice. SHAP analysis pointed to Charlson Comorbidity Index and length of stay as the most influential predictive features, but these findings were never validated against clinical evidence on what actually drives Medicare readmissions. The interpretability framework remained incomplete without systematic verification against domain knowledge.

Tang et al. [3] built multimodal spatiotemporal graph neural networks (MM-STGNN) reaching 0.79 AUROC by combining temporal chest radiographs with longitudinal EHR data. Tested on Mayo Clinic internal data (9,162 admissions) and MIMIC-IV for external validation (14,532 admissions), the model beat the LACE+ clinical standard (0.61 AUROC) by 18 points. The architecture treated patient similarity as graph structure, enabling sophisticated representation of temporal relationships. GNNExplainer provided node-level feature importance visualizations, but the authors’ careful phrasing—features “may reflect” primary diagnoses—showed the underlying problem. Without systematic comparison to clinical literature on readmission mechanisms, the explanations were hypotheses rather than verified insights.

B. XAI Application and Aggregation Strategies

Pham et al. [4] aggregated SHAP values across nine different model architectures (Logistic Regression, Random Forest, XGBoost, CatBoost, LightGBM) on Irish hospital discharge records from St. James’s Hospital in Dublin, reaching 0.70 AUROC while identifying malignant neoplasm codes, COPD (ICD-10 J44), and social determinants (Z59 codes for housing and economic problems) as top predictors. Multi-model SHAP aggregation improved explanation stability, addressing concerns about single-model variance. However, interpretation was still subjective rather than systematic. The paper explicitly requested validation frameworks to check whether the identified predictors matched peer-reviewed clinical evidence on readmission mechanisms. This gap between generating stable explanations and validating their clinical accuracy highlights the need for systematic validation frameworks.

III. THE 74% VALIDATION GAP

Two systematic reviews measured the extent of this problem, illustrated in Fig. 1.

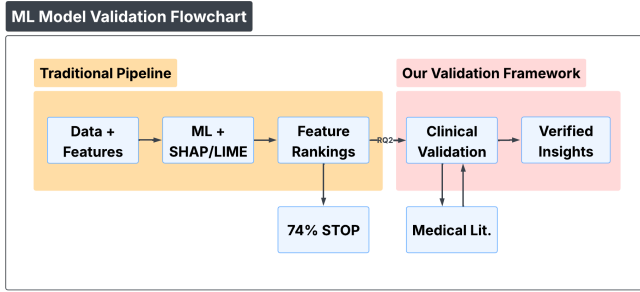


Fig. 1. XAI validation framework addressing the 74% gap. Traditional pipeline (left, yellow) generates SHAP/LIME feature rankings where most studies terminate. Proposed framework (right, pink) adds systematic clinical validation against medical literature (RQ2).

A. Systematic Review Evidence

Chaddad et al. [5] analyzed more than 450 healthcare XAI studies published between 2011 and 2022. The review revealed that SHAP became the dominant explainability method for EHR-based research, outpacing gradient-based methods like Integrated Gradients and rule-based attribution techniques. The authors categorized XAI approaches into perturbation-based (LIME and SHAP), gradient-based, and rule-based methods, evaluating their suitability for different healthcare data modalities. Critically, 74% of studies reported feature importance rankings or explanations without validating them against clinical domain knowledge or established medical evidence.

Loh et al. [6] confirmed this finding through a parallel systematic review examining XAI applications across healthcare domains from 2011 to 2022. They categorized approaches into five methodological families: dimension reduction, feature importance, attention mechanisms, knowledge distillation, and surrogate models. Their analysis confirmed SHAP and LIME as top picks for tabular EHR applications, with attention mechanisms preferred for imaging and sequential data. Only 26% of studies validated their XAI outputs against clinical domain knowledge, meaning 74% generated explanations without clinical verification. Both reviews emphasized that explanations must reflect model behavior *and* align with clinical understanding. Most studies failed to provide either form of dual validation.

Di Martino and Delmastro [7] surveyed XAI applications specifically for tabular and time-series clinical data, confirming that model-agnostic methods like SHAP and LIME work well for structured EHR analysis. The review recognized that these techniques successfully generated feature importance rankings for clinical prediction tasks, yet clinical validation strategies remained mostly unexplored. The authors identified a critical gap: while SHAP and LIME provided mathematical explanations of model decisions, no systematic frameworks existed to verify that explanations matched medical domain expertise or peer-reviewed evidence on disease mechanisms.

Hassija et al. [8] documented the fundamental accuracy-explainability tradeoff in their cognitive computation review,

demonstrating through empirical analysis that neural networks reached high predictive performance but offered low intrinsic explainability, while decision trees exhibited the opposite pattern. They categorized XAI into transparent design (intrinsically interpretable models) and post-hoc methods (SHAP, LIME for black-box interpretation). Post-hoc explanations should be clinically interpretable, but existing work skipped the validation step confirming explanations accurately reflected model reasoning and provided trustworthy clinical insights.

B. Dual XAI Frameworks and Methodology Gaps

Alnazari et al. [9] implemented dual SHAP and LIME frameworks for predicting readmissions following renal transplantation, achieving 0.837 AUROC on 588 patients with an 88.9% readmission rate. The study used SHAP for global feature importance and LIME for local instance-level explanations, identifying length of stay (38% contribution) and post-transplant systolic blood pressure (30% contribution) as primary predictors. The model was deployed as a web-based clinical decision support application at King Abdullah International Medical Research Center, demonstrating practical implementation of dual XAI approaches. Two problems remained unsolved: the study provided no validation confirming these SHAP-identified features aligned with transplant medicine literature on readmission mechanisms, and the authors did not investigate whether their class imbalance handling techniques (addressing the 88.9% positive rate) affected the stability or consistency of SHAP and LIME explanations.

Wani et al. [10] achieved clinical validation for imaging applications through radiologist review of SHAP-highlighted CT regions in lung cancer detection. Their validation approach involved domain experts verifying whether SHAP-highlighted regions corresponded to actual pathological features. This represented successful clinical validation for imaging applications, as radiologists could visually confirm explanation accuracy. However, this methodology does not transfer to tabular EHR data where no visual ground truth exists for checking feature rankings—practitioners cannot visually inspect feature importance rankings to confirm clinical validity.

IV. RESEARCH GAPS AND FUTURE DIRECTIONS

We identified four specific research opportunities from this gap analysis:

RQ1: Comparative SHAP Analysis Across Model Architectures. Pham et al. [4] specifically requested comparative explainability analysis after demonstrating that multi-model SHAP aggregation improved stability. Extending this work requires systematic comparison of SHAP outputs across fundamentally different architectures (Logistic Regression, XG-Boost, Neural Networks) to determine whether explanations converge or diverge as model complexity increases.

RQ2: Clinical Validation Frameworks for XAI Outputs. This is the primary novelty opportunity, directly tackling the 74% validation gap that Chaddad et al. [5] and Loh et al. [6] identified. Building systematic frameworks to validate SHAP and LIME results to peer-reviewed medical literature on

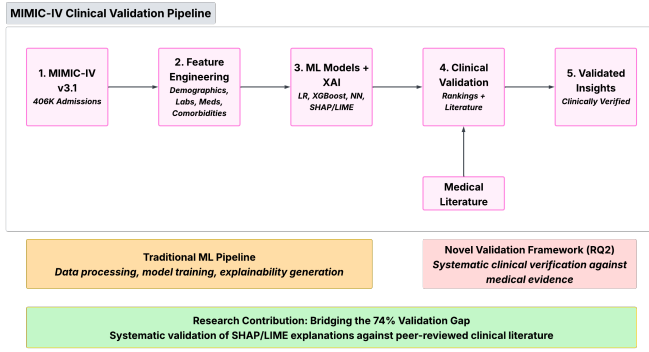


Fig. 2. MIMIC-IV v3.1 clinical validation pipeline. Traditional ML+XAI workflow (stages 1-3, yellow) processes data and generates explanations. Novel validation framework (stages 4-5, pink) implements RQ2 verification against clinical literature.

readmission risk factors would bridge the critical gap between generating explanations and checking their clinical accuracy. Validation will be measured by concordance scoring, comparing SHAP-ranked features with established readmission risk factors documented in peer-reviewed clinical literature, using a structured alignment rubric. We will apply the validation framework shown in Fig.2 to MIMIC-IV v3.1, a recent release with 406,031 hospital admissions that includes improved data quality controls, which matters especially for readmission analysis. All analysis code will be shared publicly through Jupyter notebooks to allow reproducibility and verification by other researchers. This validation would shift XAI from mathematically sound but clinically unverified outputs into trustworthy clinical insights.

RQ3: Class Imbalance Effects on Explanation Stability. Yu and Xie [1] demonstrated that class imbalance fundamentally affects prediction performance, while Alnazari et al. [9] built dual SHAP and LIME frameworks without investigating imbalance effects. No existing work examines whether SHAP and LIME explanations remain stable when applying class imbalance correction techniques like SMOTE or undersampling. This gap combines the class imbalance focus of early ensemble work with modern XAI analysis. Overall, while imbalance correction boosts predictive recall, its downstream effect on SHAP attribution stability stays unexamined in readmission prediction contexts.

RQ4: Integration of Dimensionality Reduction with Explainability. Loh et al. [6] identified dimension reduction as a distinct XAI category but noted limited integration with feature importance methods. Combining t-SNE visualization with SHAP-based feature importance could provide complementary perspectives on which features drive model decisions, representing a methodological innovation based on established approaches.

V. CONCLUSION

Machine learning models for hospital readmission prediction have reached strong predictive performance, and XAI

techniques have enabled post-hoc interpretation, but a fundamental validation gap persists. Seventy-four percent of healthcare XAI studies generate feature importance rankings without verifying their clinical accuracy against established medical knowledge, creating a significant barrier to trustworthy clinical deployment. We traced the clear progression from ensemble methods through deep architectures to XAI applications, yet systematic frameworks for validating explanations against clinical domain knowledge remain absent. Our four research opportunities—comparative SHAP analysis across architectures, development of clinical validation frameworks for XAI output, investigation of class imbalance effects on explanation stability, and integration of dimensionality reduction with explainability methods—would bridge the current divide between machine-generated explanations and clinically verified insights, advancing the development of trustworthy AI systems for healthcare decision support.

REFERENCES

- [1] K. Yu and X. Xie, “Predicting Hospital Readmission: A Joint Ensemble-Learning Model,” *IEEE J. Biomed. Health Inform.*, vol. 24, no. 2, pp. 447–456, 2020.
- [2] X. Li, S. Liu, D. Yu, Y. Zhang, and X. Liu, “Predicting 30-Day Hospital Readmission in Medicare Patients: Insights from an LSTM Deep Learning Model,” arXiv preprint arXiv:2410.17545, Oct. 2024.
- [3] S. Tang, A. Tariq, J. A. Dunnmon, U. Sharma, P. Elugunti, D. L. Rubin, B. N. Patel, and I. Banerjee, “Predicting 30-Day All-Cause Hospital Readmission Using Multimodal Spatiotemporal Graph Neural Networks,” *IEEE J. Biomed. Health Inform.*, vol. 27, no. 4, pp. 2071–2082, 2023.
- [4] M.-K. Pham, T. T. Mai, M. Crane, M. Ebiele, R. Brennan, M. E. Ward, U. Geary, N. McDonald, and M. Bezbradica, “Forecasting Patient Early Readmission from Irish Hospital Discharge Records Using Conventional Machine Learning Models,” *Diagnostics*, vol. 14, no. 21, p. 2405, 2024.
- [5] A. Chaddad, J. Peng, J. Xu, and A. Bouridane, “Survey of Explainable AI Techniques in Healthcare,” *Sensors*, vol. 23, no. 2, p. 634, 2023.
- [6] H. W. Loh, C. P. Ooi, S. Seoni, P. D. Barua, F. Molinari, and U. R. Acharya, “Application of Explainable Artificial Intelligence for Healthcare: A Systematic Review of the Last Decade (2011–2022),” *Comput. Methods Programs Biomed.*, vol. 226, p. 107161, 2022.
- [7] F. Di Martino and F. Delmastro, “Explainable AI for Clinical and Remote Health Applications: A Survey on Tabular and Time Series Data,” *Artif. Intell. Rev.*, vol. 56, no. 6, pp. 5261–5315, 2023.
- [8] V. Hassija, V. Chamola, A. Mahapatra, A. Singal, D. Goel, K. Huang, S. Scardapane, I. Spinelli, M. Mahmud, and A. Hussain, “Interpreting Black-Box Models: A Review on Explainable Artificial Intelligence,” *Cogn. Comput.*, vol. 16, pp. 45–74, 2024.
- [9] N. Alnazari, O. I. Alanazi, M. O. Alosaimi, Z. M. Alanazi, Z. M. Alhajeri, K. M. Alhussaini, A. M. Alanazi, and A. Y. Azzam, “Development of Explainable Artificial Intelligence Based Machine Learning Model for Predicting 30-Day Hospital Readmission After Renal Transplantation,” *BMC Nephrol.*, vol. 26, no. 1, p. 203, 2025.
- [10] N. A. Wani, R. Kumar, and J. Bedi, “DeepXplainer: An Interpretable Deep Learning Based Approach for Lung Cancer Detection Using Explainable Artificial Intelligence,” *Comput. Methods Programs Biomed.*, vol. 243, p. 107879, 2024.
- [11] D. Gopukumar, A. Ghoshal, and H. Zhao, “Predicting Readmission Charges Billed by Hospitals: Machine Learning Approach,” *JMIR Med. Inform.*, vol. 10, no. 8, p. e37691, Aug. 2022.